

LLM-Driven Analysis and Detection of Vulnerabilities in Filipino Credentials Using Personal Information and Score Similarity Models

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Abstract—This study introduces an LLM-driven framework for detecting vulnerabilities and cyber attacks on Filipino-based credentials through the use of synthetic personal information (PI) and score similarity models. The approach integrates several state-of-the-art open-weight Large Language Models (LLMs)—including Mistral 7B, LLaMA 3.1, Qwen 2.5, DeepSeek-R1, Phi-4 Reasoning Flash, and Qwen 2.5-Coder—to generate culturally aware password variations. A larger DeepSeek 32B model is employed to produce synthetic Filipino personal information for contextual password analysis. The framework combines LLM-based generation and similarity scoring to evaluate password strength, complexity, and vulnerability, focusing on behaviors such as password tweaking and credential reuse. This work aims to enhance password security research within the Filipino context by offering ethical, data-driven, and culturally contextualized insights into credential vulnerability.

Index Terms—

I. INTRODUCTION

A. Password Modeling

Password modeling is the practice of analyzing and simulating the creation, usage, and potential guessing of passwords by attackers. This includes examining trends in the way passwords are chosen, predicting common password formats, and creating algorithms to assess password strength or generate possible password guesses. [1]

Password similarity scoring is a related concept to password modeling that measures the likeness of different passwords. This score is used to detect small variations, such as those found in credential tweaking attacks, where an attacker makes slight changes to a known password in order to evade security defenses. One of the most commonly used metrics to assess password similarity is the Levenshtein distance [2]. This metric is used to measure the difference between two strings by counting the minimum number of single-character edits required to transform one string into the other. These edits can include insertions, deletions, or substitutions. It is commonly used in applications such as spell checking, DNA sequencing, and natural language processing to assess the similarity between words or sequences. [3]

Password guessing attacks aim to infiltrate user accounts by predicting or acquiring passwords. These attacks range

from targeted strategies, which focus on specific individuals or organizations using personal information, to untargeted approaches, which rely on the probability that some users will employ common or previously compromised passwords. Offline versus Online Attacks Offline attacks involve cracking password hashes on a system separate from the target, allowing for rapid and undetected attempts. Online attacks involve guessing passwords directly against an active system, where the attacker is restricted by the network speed and security measures of the system. In both scenarios, the adversary's goal is to successfully crack passwords before exhausting their budget (query budget), whether it is the number of allowed attempts in an online service or the computing resources available for offline guessing.

Compromised Credential Checking (C3) services play a vital role in enhancing the security of authentication systems by identifying compromised or weak credentials before they can be exploited by attackers. These services typically operate by comparing user-provided credentials against large databases of known breached credentials, alerting users and administrators when a match is found. One of the most widely recognized C3 services is "Have I Been Pwned" (HIBP), developed by Troy Hunt. HIBP allows users to check if their email addresses or passwords have appeared in any known data breach. It also provides an API that organizations can integrate into their security infrastructure to automate the detection of compromised credentials. [4]

The Philippines showcases unique password behaviors, as evidenced by common passwords like "P0weranger," "iloveyou," and "alien123," [19] reflecting cultural tendencies towards affection and pop culture. With the rapid digitalization of the country, driven by increasing internet access and smartphone usage, there is a growing need for stronger cybersecurity awareness and practices. As demand for cybersecurity rises, Filipino users are likely to shift towards more secure password practices, including using complex passwords and multi-factor authentication [20], [21]. Training models with Filipino credentials can reduce noise and enhance data quality by focusing on relevant, localized information. Moreover, training with massive datasets can be inefficient due to time and computational costs, making a more targeted, curated approach more effective. Studies show that fine-tuning large language models (LLMs) benefits from high-quality, diverse datasets, emphasizing the importance of curation over sheer quantity. LLMs are better attuned to specific regional

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behaviors like those seen in Filipino password practices [22].

With the increasing frequency of cyber attacks and credential leaks, analyzing password security has become a critical focus in cybersecurity research. Many global studies have examined password reuse, credential stuffing, and cultural password tendencies. However, limited work exists on the Filipino context, where password selection behaviors often integrate personal and cultural elements. These traits can make Filipino-based credentials more susceptible to targeted attacks.

This study introduces a novel LLM-driven framework that leverages synthetic Filipino personal information (PI) and score similarity models to evaluate password vulnerabilities. The goal is to assess the extent to which cultural and personal identifiers influence password strength, while maintaining ethical research standards by using synthetic data rather than real leaked credentials.

The proposed framework utilizes advanced LLMs to generate, evaluate, and compare culturally relevant password variations, providing insights into Filipino credential security patterns and supporting the creation of practical password security guidelines.

B. Significance of the Study

The relevance of this study lies in its ability to bridge the gap between local cybersecurity needs and global attack trends. By focusing on Filipino-based credentials and applying score similarity models, the study offers a proactive approach to detecting and mitigating credential-based attacks. This research is crucial for improving cybersecurity defenses in the Philippines, where the use of digital platforms is growing rapidly but cyber awareness and defenses may still be lagging. Ultimately, this study will contribute to a more secure online environment, both locally and globally.

C. Objectives of the Study

This study aims to explore password security within the context of Filipino users by analyzing real-world credentials and simulating password variations through Large Language Models. Specifically, the objectives are to:

- 1) analyze credential selection behavior, focusing on how Filipino users create and reuse credentials, including common password composition patterns and usage behaviors;
- 2) identify weaknesses in Filipino credentials, emphasizing vulnerabilities related to password tweaking practices and the inclusion of personal information (PI);
- 3) evaluate LLM-generated password variations by comparing them against real-world breached password pairs in terms of strength, similarity, and vulnerability using score similarity models;
- 4) evaluate the developed website in terms of its accuracy and usability; and

- 5) recommend practical and culturally relevant guidelines for developing stronger, more secure, and less predictable Filipino credentials, while promoting user awareness of password security.

D. Scope and Limitations

This study focuses on Filipino-based credentials, specifically examining vulnerabilities from credential stuffing and tweaking attacks. It utilizes existing Large Language Models (LLMs)—the DeepSeek 32B model for generating synthetic personal information (PI) and smaller models (1.5B–7B) for password variation generation without requiring custom training. Only datasets containing Filipino and English credentials under the (.ph) domain are used. Ethical safeguards include strict data filtering and the exclusive use of synthetic identities to prevent real credential exposure. Prompt engineering techniques are applied to ensure consistent, realistic, and culturally relevant password generation.

II. REVIEW OF RELATED LITERATURE

Password security literature is extensive which underscores its critical importance within cybersecurity research. This section reviews the key developments in password models and guessing attacks, password similarity score, and compromised credential checking (C3) protocols.

A. 2.1 Compromised Credential Checking (C3) Services

Compromised Credential Checking (C3) services are used in authentication systems by identifying compromised or weak credentials. Having weak credentials will make a user vulnerable to cyber attacks. One of the most widely used C3 service is “Have I Been Pwned” (HIBP) by Troy Hunt. It allows users to check if their credentials are previously included in data breaches. This approach have some risk to the user since some might actually use their own credentials and it might get compromised. Another flaw is that there is a weak encryption method used by website in passwords. Although its API is still widely integrated into security of many websites. Other notable C3 services include:

- Google Password Checkup (GPC) [5]: Integrated into Google’s ecosystem, this service alerts users if their saved passwords have been compromised in any data breaches. It provides proactive warnings to encourage users to change vulnerable passwords.
- Firefox Monitor [6]: Developed by Mozilla, Firefox Monitor informs users if their email addresses have been involved in any known breaches. It also offers recommendations for improving password security.
- Microsoft AccountGuard [7]: This service provides additional security measures for high-risk users, such as those in political campaigns or human rights organizations. It includes monitoring for compromised credentials and guidance on securing accounts.
- 1Password Watchtower [8]: Integrated into the 1Password password manager, Watchtower continuously scans for breached passwords and alerts users to take corrective action.

- Cloudflare’s Might I Get Pwned (MIGP) [10]: MIGP is a service provided by Cloudflare that allows users to securely check if their credentials have been exposed in data breaches. It leverages k-anonymity and cryptographic hashing techniques to ensure user privacy while providing accurate breach detection.

Each of these services has contributed to raising awareness about the importance of credential security and has provided users with tools to protect their digital identities against password guessing attacks. Recent advancements in C3 protocols have further enhanced security and efficiency. For example:

- Frequency Smoothing Bucketization (FSB) scheme [11] introduces a method to distribute credentials into buckets more evenly, reducing the risk of attackers identifying frequently used passwords. This approach enhances privacy by making it more challenging for an attacker to deduce the popularity of specific credentials.

- Oblivious Pseudorandom Function (OPRF) protocol [4] plays a pivotal role in enhancing security, particularly in the context of password authentication and privacy preservation. It enables a two-party protocol where a client can receive the output of a Pseudorandom Function (PRF) from a server without revealing the input or learning about the server’s PRF key. This ensures that sensitive data, such as passwords, are processed in a manner where neither party gains access to information beyond their requisite knowledge. In the domain of password checking, the OPRF protocol contributes to the secure and private computation of password similarity scores, crucial for detecting credential tweaking without compromising user privacy.

B. 2.2 Password Guessing Attacks

Password modeling has evolved significantly since its inception as a cryptanalytic challenge. Initially focused on brute-force and dictionary attacks, the field has expanded to incorporate natural language processing (NLP) and machine learning (ML) techniques. This shift recognizes passwords as linguistic constructs with analyzable patterns[4]. Modern password modeling serves multiple purposes such as developing advanced password guessers, understanding password reuse behaviors, assessing password guessing attack risks, and creating sophisticated password strength meters. A variety of password generation models have been introduced over time, advancing the development of password guessers, understanding password reuse behavior, evaluating the risks of password guessing attacks, and creating sophisticated password strength meters.

C. 2.3 Password Modeling

Password modeling refers to the process of understanding and simulating how passwords are created, used, and potentially guessed by attackers. This involves analyzing patterns in password selection, predicting likely password structures, and developing algorithms that can evaluate password strength or generate potential password guesses. There are many factors that can affect how a password is created such as demographic of users and website requirements.

D. 2.4 Fine-tuning of Large Language Models

Large Language Models are transformers that are fed data to help in the understanding of behavior and creation of more data. It is widely used by GPTs and have become their foundation. LLMs are a great way to minimize work and accelerate progress through fine-tuning existing ones. Fine-tuning involves the data processing part and changing of variables and weights within the system[18]. Adding in more data does not always mean that the LLM will have greater capabilities. Having lots of data will increase the chance of getting noise. Quality data is much more appreciated. A way of increasing the quality is filtration of the data that is going to be fed.

E. 2.5 Password Similarity Score

While existing methods for evaluating password similarity, such as edit distance and zxcvbn score, offer valuable insights into structural similarities between passwords, these metrics may not fully capture the likelihood of password generation patterns, particularly in the context of credential tweaking and credential stuffing. To address the gap a combination of different similarity scores can be used. Usage of edit distance, similarity ratio and zxcvbn Score together in one formula can be implemented. Another score similarity is cosine similarity. Cosine similarity is a metric used to determine the similarity between two non-zero vectors by measuring the cosine of the angle between them. It indicates how closely aligned the two vectors are, with a value of 1 meaning identical, 0 indicating no similarity, and -1 signifying completely opposite directions. This measure is commonly used in text analysis, information retrieval, and machine learning to compare the similarity of documents or data points [18].

F. 2.6 PassGAN and Its Evolution to PAGPass and PAGPass-GPT

PassGAN was first introduced by Hitaj et al. (2019) as a Generative Adversarial Network (GAN)-based approach to password generation. Unlike traditional rule-based password guessers, PassGAN uses a generator and discriminator network to model the distribution of real passwords and produce realistic password guesses. Its strength lies in the ability to generate a large number of plausible passwords by learning patterns from existing datasets without explicit rules. However, PassGAN’s performance is limited by the scope of its training data, which primarily reflects global or English-centric password datasets, and lacks cultural or regional specificity [23].

To address the need for localized password generation, PAGPass (Password-Aware GAN Pass) was developed. PAGPass extends PassGAN by incorporating domain-specific and cultural patterns into the training process. This allows the model to generate passwords that reflect local tendencies, such as Filipino users’ integration of names, dates, or pop culture references. By doing so, PAGPass improves the relevance and accuracy of generated passwords when applied to region-specific datasets, making it more effective in detecting and modeling password vulnerabilities within a given context [24].

Building upon PAGPass, PAGPassGPT integrates Large Language Models (LLMs) such as GPT-style transformers into the GAN framework. This hybrid approach leverages the contextual and semantic understanding of LLMs to produce passwords that not only mimic observed patterns but also account for linguistic and personal information cues. PAGPassGPT supports similarity scoring and password variation generation, making it particularly suitable for analyzing culturally contextualized credentials. By combining GANs with LLMs, PAGPassGPT represents a significant advancement in password modeling, allowing researchers to evaluate vulnerabilities with higher fidelity and ethical synthetic data [?].

G. LLM-Based Evaluation of Password Strength

Traditional password strength meters, such as zxcvbn, rely on rule-based scoring that evaluates factors like length, character diversity, dictionary words, and common patterns (e.g., repetitions, keyboard sequences, and leetspeak). While these methods are fast and deterministic, they lack adaptability to emerging password trends or cultural nuances unless manually updated. Prior studies have shown that incorporating language-specific dictionaries can improve accuracy, as seen in versions tailored for Korean and Lithuanian users.

Recently, Large Language Models (LLMs) have emerged as flexible tools for password evaluation. Unlike fixed-rule systems, LLM-based evaluation leverages the models' broad linguistic and contextual understanding to assess password strength dynamically, without additional training or algorithmic modification. Through prompt-driven evaluation, LLMs can recognize deeper cultural and contextual cues (e.g., Summer2024!, Juan1997!) and adapt to evolving password creation behaviors.

The study “A Novel Approach to Password Strength Evaluation Using ChatGPT-Based Prompt Metrics” demonstrated this concept using GPT-4 with prompt-based metrics such as *Complexity*, *Memorability*, and *Personal Information (PI) Inclusion*. This method provides contextual awareness and cultural sensitivity, enabling more accurate and human-like assessments of password vulnerability compared to traditional rule-based systems.

III. MATERIALS AND METHODS

A. Data Gathering and Filtration

The foundation of this study is based on publicly available breached datasets, primarily the Compilation of Breached Data (COMB) and 4iQ collections. These datasets undergo a multi-stage filtration process to ensure data quality and cultural relevance:

- 1) **Email Filtering:** Only email addresses associated with the .ph domain are considered. For major providers (e.g., Gmail, Outlook, Yahoo, Hotmail, ProtonMail), a simplified regular expression $([a-zA-Z0-9._\%]+)$ is applied to validate the local-part. For other domains, an RFC 5322-compliant regex ensures proper formatting. Disposable and IANA-reserved domains are excluded.
- 2) **Password Filtering:** Passwords are filtered according to NIST SP 800-63B guidelines, requiring a minimum

length of eight and a maximum of sixty-four characters. Only printable ASCII and normalized Unicode characters are retained, with non-printable symbols removed.

- 3) **Cultural Relevance:** The final filtration step ensures inclusion of passwords containing common Filipino names, words, or expressions, reflecting cultural and linguistic traits. This enhances contextual accuracy and local relevance. The resulting dataset includes cleaned email-password pairs and a list of unique passwords, accompanied by a summary report detailing input counts, valid credentials, unique users, and unique passwords.

B. Synthetic Identity and Personal Information (PI) Dictionary Generation

To ethically assess the influence of Personal Information (PI) on password vulnerability without using real user data, a synthetic identity generator will be implemented. This generator employs existing LLMs (DeepSeek 32B) through structured prompt engineering to produce realistic but fictitious identities. The generated profiles include fields such as full name, birth year, gender, city, occupation, hobbies, and pet names, with distributions designed to reflect real-world Filipino demographic patterns. The resulting synthetic data will serve as the foundation for constructing a PI dictionary for password strength and similarity evaluation.

C. LLMs Utilized for Password Variation Generation

To generate realistic and culturally relevant password variations, this study employs several state-of-the-art Large Language Models (LLMs). These models were selected based on their open availability, multilingual reasoning capabilities, and balance between computational efficiency and contextual understanding. The following models are utilized:

- **Mistral 7B** — a high-efficiency model optimized for general-purpose text generation and reasoning, suitable for structured generation tasks.
- **Qwen 2.5 7B** — a multilingual LLM developed by Alibaba Cloud, capable of context-aware generation with strong handling of localized linguistic cues.
- **Gemma 3 4B** — a lightweight, open-weight model designed for efficient reasoning and text synthesis while maintaining compositional consistency.
- **LLaMA 3.1 8B** — Meta's latest general-purpose model known for balanced reasoning performance and multilingual adaptability.
- **Phi 3 14B** — a reasoning-tuned model optimized for instruction following and analytical coherence, used for comparative password strength evaluation.

Each model contributes distinct linguistic and reasoning capabilities to the password generation framework. By combining their outputs, the system produces diverse, high-fidelity password variations that reflect Filipino linguistic patterns and cultural nuances while maintaining ethical and synthetic data boundaries.

D. Prompt Engineering and Design

Prompt engineering guided the behavior of large language models (LLMs) to ensure outputs were culturally consistent, ethically generated, and aligned with password generation objectives. Two structured templates were designed for this process.

1) *Synthetic PI Generation*: The model was instructed to create plausible yet synthetic personal information based on provided credentials. The pseudocode below outlines the generation logic:

[1] Input: Email, Password Output: JSON object with {email, password, full_name, birthdate, phone_number, address, gender, reasoning} Ensure:

- Phone number follows PH format ("09" + 9 digits)
- Birthdate plausible based on password hints
- Reasoning field briefly explains relationship between PI and credentials

Generate realistic PI consistent with email/password context
Return single validated JSON object

2) *Password Variation Generation*: The second template generates password variants derived from the enriched credentials. The pseudocode defines the generation workflow:

[1] Input: Base password + user info (optional) Define rules:

- Maintain human-like tweaks (year ± 1 , leetspeak, prefixes/suffixes)
- Include at least one uppercase, lowercase, digit, and symbol
- Avoid reusing substrings > 4 characters from base password

Generate N structurally consistent variants Output: JSON array of unique password strings

Prompts are applied uniformly across all models to ensure comparability.

3) *Hybrid Similarity Scoring (HSS)*: To quantify password similarity and detect potential reuse or modification, this study adopts the Hybrid Similarity Scoring (HSS) model proposed by Ajes *et al.* [24]. The HSS metric combines the strengths of Cosine Similarity (*CosSim*) and Damerau–Levenshtein Similarity (*DLSim*) to evaluate both character-level statistical resemblance and structural edits between two passwords.

$$HSS = (w_1 \times CosSim(pw_1, pw_2)) + (w_2 \times DLSim(pw_1, pw_2)) \quad (1)$$

where:

- pw_1 and pw_2 represent the two passwords being compared;
- *CosSim* captures statistical similarities in character frequency, detecting variations such as capitalization changes, leetspeak substitutions, and repeated character patterns;
- *DLSim*, normalized by the length of the longer password, measures structural modifications including character insertions, deletions, swaps, and substitutions;
- w_1 and w_2 are adjustable weight parameters, each initially set to 0.5 to balance the influence of both metrics.

This hybrid scoring method effectively captures both semantic and structural password similarities, providing a more

comprehensive similarity measure compared to traditional single-metric approaches.

E. Implementation Flow

The implementation follows a data-driven pipeline designed to model culturally contextual password behavior through LLM-generated synthesis and evaluation. The process, illustrated in Figure 1, consists of these main stages:

- 1) **Cultural and Credential Collection**: Filipino-related credentials were extracted from breached datasets (COMB and 4iQ) and analyzed for cultural language patterns and password composition. These filtered entries formed the foundation for generating realistic synthetic identities.
- 2) **Synthetic Data Generation**: DeepSeek-R1-32B was employed to produce synthetic personal information (PI) such as name, birthdate, phone, and address, ensuring cultural and contextual coherence with the original Filipino credentials. These enriched inputs were structured for model processing.
- 3) **LLM-Based Credential Generation**: The prepared PI-credential pairs were passed to reasoning-oriented LLMs (DeepSeek-R1, Phi-4-Reasoning, and Qwen-QwQ). Each model generated two outputs: (a) composition-aware base passwords, and (b) culturally realistic password variants derived from inferred patterns.
- 4) **Evaluation and Scoring**: Generated outputs were evaluated through two main dimensions: (a) password strength using *zxcvbn-ts*, and (b) similarity using a Hybrid Similarity Score (HSS) combining Damerau–Levenshtein distance for structural proximity and cosine similarity for semantic likeness.
- 5) **Integrated Cultural Analysis**: Final results combined strength and similarity assessments to identify trends in password complexity, cultural vulnerability, and model performance. Insights derived support the creation of culturally aware password-strength evaluation systems.

This streamlined workflow ensures that password vulnerability is analyzed ethically and contextually through a combination of real-world data, synthetic identity generation, and advanced LLM-based reasoning.

F. Development of the Password Strength and Similarity Website

A simple web-based tool was developed to evaluate password strength and similarity. The application was created using modern front-end frameworks — React and Vue — to ensure a responsive and interactive user interface. The tool allows users to enter their email and current password, with optional user inputs such as names, dates, or phrases to detect similar patterns. After entering a new password, the system analyzes it using the *zxcvbn* algorithm to produce a strength score ranging from 0 (very weak) to 4 (very strong). It also calculates a similarity score between the current and new passwords, where 0 represents no similarity and 1 indicates identical passwords.

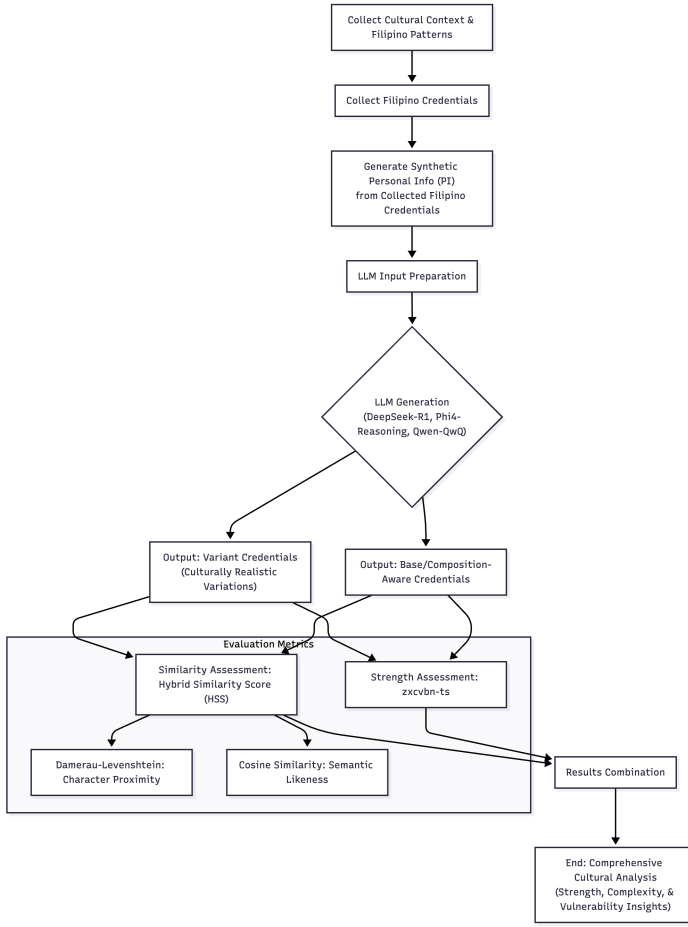


Fig. 1. Static website flow of the proposed framework.

G. Evaluation of Website

To evaluate the effectiveness, accuracy and usability of the website, users will be asked to answer a Standardized User Experience Percentile Rank Questionnaire (SUPR-Q) [16].

- 1) The website is easy to use.
- 2) It is easy to navigate within the website.
- 3) How likely are you to recommend this website to a
- 4) I will likely return to the website in the future.
- 5) The website has a clean and simple presentation.
- 6) I am able to find what I need quickly.
- 7) I can count on the information I get on this website.
- 8) I enjoy using the website. friend or colleague?

All items use a 5-point scale, with 1 representing "strongly disagree" and 5 representing "strongly agree."

IV. RESULTS AND DISCUSSION

A. Data Gathering and Filtration

The initial dataset comprised a total of **884,554,596** credential entries extracted from the *Compilation of Many Breaches (COMB)* and *4iQ* databases. To ensure both data integrity and cultural relevance, only credentials associated with the .ph domain were retained. Following the multi-stage filtration process described in the methodology, the resulting dataset metrics are summarized in Table I.

TABLE I
SUMMARY OF FILIPINO CREDENTIAL FILTERING FROM BREACHED DATASETS

Metric	Count / Value
Total entries (raw)	884,554,596
After filtration (valid .ph emails)	265,668
Invalid entries removed	884,288,928
Unique passwords identified	212,253
Minimum password length	7 characters
Maximum password length	64 characters
Primary sources	COMB, 4iQ

The filtration process significantly reduced the dataset from nearly **885 million** raw entries to approximately **266 thousand** culturally relevant Filipino credentials. This reduction of over **99.9%** highlights the rarity of Philippine-related credentials within large global breach corpora, emphasizing the necessity of localized password analysis.

The remaining **212,253 unique passwords** provide a sufficiently diverse sample for studying *credential reuse*, *password composition behavior*, and *cultural naming patterns*. The observed minimum and maximum password lengths (7–64 characters) align with the *NIST SP 800-63B* digital identity guidelines, ensuring compliance with international password policy standards.

This refined dataset serves as the foundation for the subsequent stages of analysis, including *synthetic personal information generation* and *LLM-based password variant modeling*.

B. Password Composition Analysis

Following the dataset filtration, an exploratory analysis was conducted to examine the structural composition of Filipino passwords. Figure 2 and Figure 3 present the overall length distribution and character-type composition, respectively.

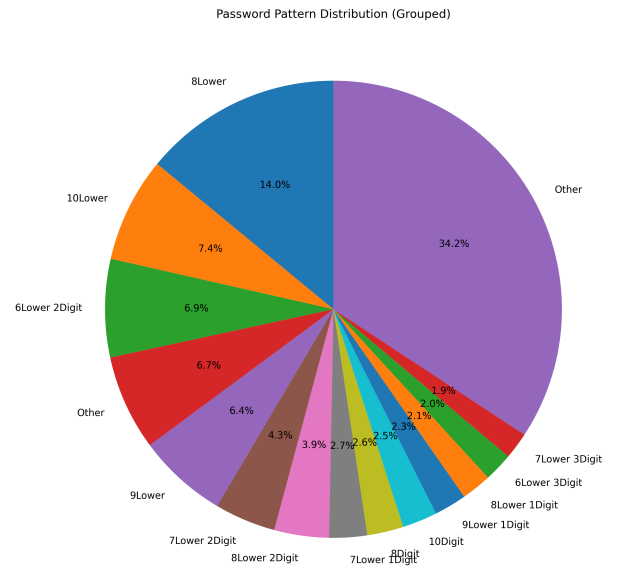


Fig. 2. Password length distribution among Filipino users.

The password composition analysis reveals that most users rely on short, lowercase, and predictable word-based passwords,

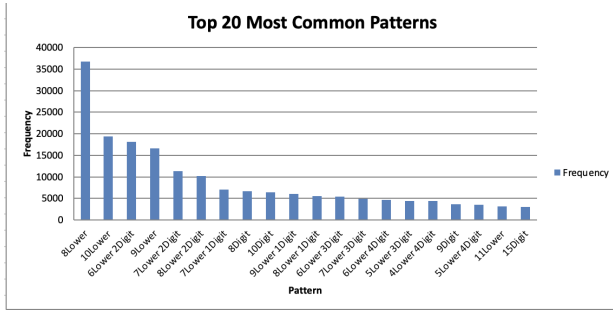


Fig. 3. Character-type composition of Filipino passwords (letters, digits, symbols, case usage).

typically ranging from **6 to 10 characters** and containing only **1–2 digits**. Purely numeric passwords are also prevalent, suggesting the frequent use of *personal information* such as birth years, phone numbers, or significant dates.

Only a small fraction of users incorporate symbols or upper-case letters, indicating limited adherence to strong password composition guidelines. This pattern reflects **low entropy and high predictability**, as users tend to prioritize *convenience and memorability* over security robustness.

Such findings are consistent with previous research on global password behaviors, reaffirming that Filipino users exhibit similar tendencies in balancing usability with security risk. The results underscore the importance of culturally contextual password-strength evaluation and education initiatives.

C. Common Substring Analysis

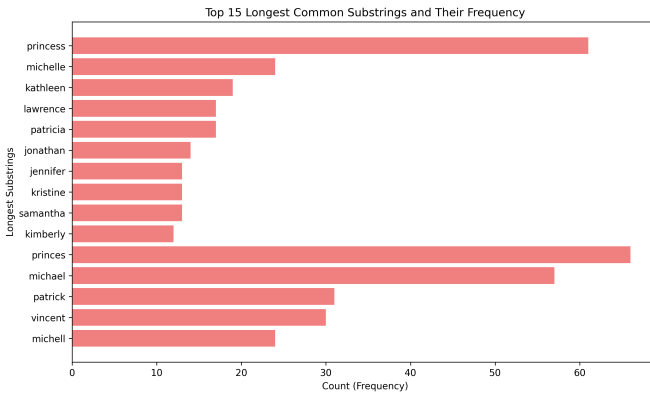


Fig. 4. Most frequent substrings appearing across Filipino passwords.

The common substring analysis shows that many passwords share short, familiar fragments such as names, emotional words, and simple numbers (e.g., "juan", "love", "123"). These repeated patterns indicate that users frequently construct passwords from memorable words or culturally significant terms rather than random character sequences. Consequently, password diversity and entropy are significantly reduced, increasing susceptibility to guessing and brute-force attacks.

D. Password Strength Analysis

The password strength analysis compares baseline zxcvbn-derived scores before and after integrating a Filipino wordlist

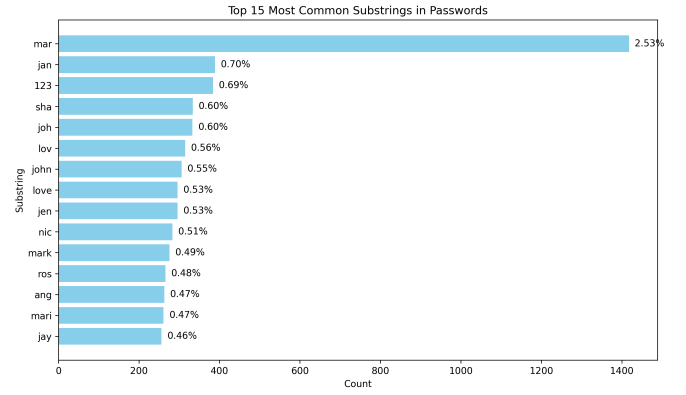


Fig. 5. Heatmap of substring frequency and co-occurrence in credentials.

TABLE II
PASSWORD STRENGTH ANALYSIS SUMMARY

Metric	Without Filipino Dictionary	With Filipino Dictionary
Average Score	2.67	1.51
Average Guesses	1.14×10^{13}	2.86×10^8

into the password-cracking dictionary. Results show a significant decline in average password strength, from 2.67 to 1.51 and a corresponding reduction in estimated guesses required for compromise, from 1.14×10^{13} to 2.86×10^8 .

Out of 265,000 analyzed credentials, only around 3,200 (1.2%) exhibited a score change, indicating that most Filipino users rely on alphanumeric or English-based passwords. However, the subset that includes Filipino words experienced a sharp drop in strength, revealing heightened vulnerability once local linguistic patterns were considered.

This suggests that integrating cultural and linguistic awareness into password strength evaluation is crucial for realistic security assessments in localized contexts.

E. Synthetic Personal Information (PI) Generation

Scope. To focus on credentials most affected by linguistic context, PI generation was restricted to the subset of **3,200** passwords whose strength score changed after integrating the Filipino wordlist. These records were used as seeds to produce synthetic personal information for targeted vulnerability experiments.

Model and inputs. I used the DeepSeek-R1 (32B) model to generate realistic but fictitious personal records. Input to the generator consisted of sanitized credential structure (local-part of email and non-identifying password cues) only, no real identities were used as seeds. Each synthetic record contains: email, full_name, birthdate, phone_number, address, gender, and a short reasoning field explaining the relation between the credential cue and the generated PI.

Validation and constraints. Each generated PI record was validated using simple, deterministic checks to ensure realism:

- Phone numbers conform to the Philippine mobile format (09XXXXXXXXXX) and pass syntactic checks.
- Birthdates are constrained to plausible human ages consistent with numeric cues present in credentials.

TABLE III
SCORE DISTRIBUTION BEFORE AND AFTER APPLYING FILIPINO
DICTIONARY

Score	Without Dictionary	With Dictionary
0	0	129
1	96	1566
2	1174	1237
3	1597	257
4	322	0

TABLE IV
SYNTHETIC PI GENERATION FOR THE VULNERABLE SUBSET

Metric	Value
Credentials processed	3,200
Synthetic records generated	3,200
Validated PH-format phone numbers	3,200
Records with plausible reasoning notes	3,200
Unique generated full names	(see repository / supplementary file)

- Addresses follow Philippine-like structure (barangay/city/province) but are fictionalized.
- Reasoning fields are short (1–2 sentences) and describe why the model associated a specific PI token with the input credential (e.g., numeric hint, local-part match).

Representative examples. Two illustrative examples of generated synthetic PI records are shown in Figures 6 and 7.

```
{
  "email": "01@yahoo.com.ph",
  "base_password": "jimwell04",
  "variants": [
    {
      "variant": "JImWell1994!",
      "damerau_levenshtein": 0.41666666666666663,
      "cosine_similarity": 0.5276448530110863
    },
    {
      "variant": "jimwell777$",
      "damerau_levenshtein": 0.6363636363636364,
      "cosine_similarity": 0.6225430174794672
    },
    {
      "variant": "jimwell2003@",
      "damerau_levenshtein": 0.6666666666666667,
      "cosine_similarity": 0.82915619758885
    },
    {
      "variant": "jimwell90!",
      "damerau_levenshtein": 0.8,
      "cosine_similarity": 0.8703882797784892
    },
    {
      "variant": "jimwell04#$",
      "damerau_levenshtein": 0.8181818181818181,
      "cosine_similarity": 0.9198662110078
    }
  ]
}
```

Fig. 6. Example 1 — Synthetic PI record generated from sanitized credential input.

Usage. The 3,200 synthetic records were used to:

- 1) build a PI dictionary for `zxcvbn-ts` evaluations (to

```
{
  "email": "jubeix_lharry11@yahoo.com.ph",
  "base_password": "gwapokoh123",
  "variants": [
    {
      "variant": "Harrington1998!",
      "damerau_levenshtein": 0.13333333333333333,
      "cosine_similarity": 0.30261376633440124
    },
    {
      "variant": "gwapo_l123",
      "damerau_levenshtein": 0.7272727272727273,
      "cosine_similarity": 0.7893522173763263
    },
    {
      "variant": "Beauxis!777",
      "damerau_levenshtein": 0.09090909090909094,
      "cosine_similarity": 0.06726727939963126
    },
    {
      "variant": "lharry_2005@",
      "damerau_levenshtein": 0.08333333333333337,
      "cosine_similarity": 0.20801257358446093
    },
    {
      "variant": "gwapokoh99",
      "damerau_levenshtein": 0.7272727272727273,
      "cosine_similarity": 0.7412493166611013
    }
  ]
}
```

Fig. 7. Example 2 — Model-generated reasoning and contextual PI synthesis.

quantify the effect of PI-aware dictionaries on password strength), and

- 2) seed prompts for downstream LLM-based password-variation generation and similarity experiments.

F. LLM Simulation of Filipino Password Vulnerability

This section evaluates how Large Language Models (LLMs) simulate Filipino users' password modification behavior, particularly when personal information (PI) is used to create password variations. The experiment aimed to determine whether LLMs could replicate the human tendency to “tweak” known passwords, a behavior that increases vulnerability to targeted attacks.

1) *Principle and Rationale:* The LLMs were prompted to generate password variants based on synthetic personal information (PI), simulating how users incorporate personal cues such as names, birthdays, and numeric patterns. The **Score Drop** (Δ) metric represents the reduction in password security when user-specific PI is introduced. A higher Δ indicates greater susceptibility to tweaking attacks.

2) *Performance Overview:* Table V summarizes the comparative results across all models, showing average strength scores and similarity metrics. Mistral:7B exhibited the highest vulnerability ($\Delta = 0.264$), indicating that it most closely replicated human-like password tweaking behaviors. In contrast, LLaMA 3.1:8B and Qwen 2.5:7B showed lower Δ values, suggesting they were less sensitive to PI correlations.

TABLE V
AVERAGE SCORES AND SIMILARITY METRICS PER MODEL

Model	No PI	With PI	Δ	Similarity
Phi-3:14B	3.34	3.18	0.16	0.48
Mistral-7B	3.06	2.80	0.26	0.43
Gemma-3:4B	2.63	2.48	0.15	0.53
LLaMA-3.1:8B	2.45	2.39	0.06	0.35
Qwen-2.5:7B	2.38	2.34	0.04	0.22

3) *Interpretation of Results:* **Mistral:7B** achieved the highest Score Drop ($\Delta = 0.26$), signifying the strongest correlation between generated passwords and user PI. This indicates that the model most accurately emulated human password “tweaking” behavior—making it the most effective simulator of human vulnerability.

Phi-3:14B, while demonstrating moderate Δ , maintained the highest average security score (3.18), implying that its reasoning-focused architecture favors complex, less predictable passwords.

Conversely, **LLaMA-3.1:8B** and **Qwen-2.5:7B** yielded the smallest Δ values (0.06 and 0.04, respectively), showing minimal sensitivity to personal information and thus low vulnerability to PI-driven predictability.

The analysis confirms that Filipino users who modify previously leaked passwords—especially by integrating PI such as names or dates—are highly vulnerable to tweaking attacks. The LLMs, acting as *predictive attackers*, successfully inferred and replicated these weak password-creation habits in a zero-shot context.

The relationship between higher Δ values and PI usage validates the hypothesis that culturally conditioned password practices (e.g., using nicknames, Tagalog terms, or birthdays) are both predictable and reproducible by modern LLMs. Hence, models like Mistral:7B demonstrate how AI systems can emulate and exploit the same weaknesses prevalent in real-world Filipino password culture.

V. DISCUSSION

This section presents the key findings derived from the experiments, aligned with the four research objectives. Each finding integrates both quantitative and qualitative insights from the password dataset, similarity scoring, and LLM-based simulations.

TABLE VI
AVERAGE SECURITY SCORES AND VULNERABILITY METRICS ACROSS LLMs

Model Name	Score (No PI)	Score (With PI)	Score Drop (Δ)
Phi-3:14b	3.339	3.178	0.161
Mistral:7b	3.059	2.795	0.264
Gemma 3:4b	2.632	2.479	0.152
Llama 3.1:8b	2.453	2.391	0.062
Qwen 2.5:7b	2.382	2.340	0.042

1. Credential Selection Behavior. The analysis revealed that Filipino users commonly create short, word-based, and easily memorable passwords (6–10 characters) often containing digits that reflect personal cues such as birthdates or phone

numbers. The prevalence of lowercase-only and minimal-symbol passwords highlights a trade-off between convenience and security. This behavior mirrors the global trend of weak password reuse but is accentuated in the Filipino context due to cultural naming and numeric habits.

2. Weaknesses and Vulnerabilities. The integration of a Filipino dictionary into strength evaluation caused a notable decrease in average password scores, from 2.67 to 1.51 for affected samples, indicating that personal and linguistic cues substantially weaken security. Additionally, the subset of 3,200 credentials that exhibited score drops demonstrated high susceptibility to tweaking attacks. This finding confirms that passwords derived from PI (names, dates, or local words) are structurally predictable and easier to compromise.

3. LLM-Based Evaluation of Password Variations. Using LLMs such as Mistral:7b, Phi-3:14b, and Gemma 3:4b, password variants were generated and compared against real-world breached pairs using the Hybrid Similarity Score (HSS). Among all models, Mistral:7b showed the largest average score drop ($\Delta = 0.264$), indicating strong alignment between the model’s generated variants and realistic human password-tweaking behavior. Phi-3:14b, despite moderate PI influence ($\Delta = 0.161$), maintained higher overall strength, suggesting it generates more complex and less predictable passwords.

The developed website, *Password Strength and Similarity Checker*, allows users to evaluate their passwords’ security by comparing their current and proposed new passwords. It provides a zxcvbn strength score (ranging from 0 to 4), a similarity score (0 = no similarity, 1 = identical), and practical suggestions to improve password quality.

As shown in Figure 8, the interface was designed using React and Vue frameworks, offering a simple and user-friendly layout. Users can input their current password, email, and optional personal keywords such as names or dates, followed by their proposed new password.

Fig. 8. Input interface of the Password Strength and Similarity Checker. Users enter their current and new passwords along with optional personal keywords.

Figure 9 shows the analysis result section of the website. After submission, the tool computes and displays the zxcvbn

score, similarity score, and textual feedback for password improvement.

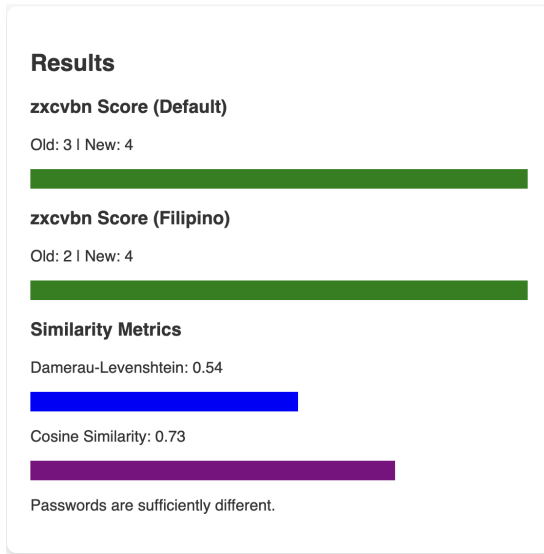


Fig. 9. Result display showing password strength, similarity, and suggestions for improvement.

VI. CONCLUSION AND FUTURE WORK

This study analyzed how Filipino users create and reuse credentials, identified vulnerabilities in password construction, and evaluated the capacity of modern Large Language Models (LLMs) to replicate culturally influenced password behaviors. The results demonstrate that password composition among Filipino users remains dominated by short, lowercase, and semantically predictable structures, often embedding personally identifiable information (PI) such as names, dates, or common Tagalog terms. These characteristics contribute to low entropy and high susceptibility to credential stuffing and tweaking attacks.

Through the generation of 3,200 synthetic PI-enriched records and subsequent password variation experiments using five state-of-the-art LLMs (Mistral:7B, Qwen 2.5:7B, Gemma 3:4B, LLaMA 3.1:8B, and Phi 3:14B), the study found that **Mistral:7B exhibited the highest score drop** ($\Delta = 0.26$), indicating the strongest tendency to replicate human-like tweaking behavior and thus simulate the most vulnerable pattern. Conversely, **Phi 3:14B achieved the best overall security score** (3.18) while maintaining moderate PII correlation, suggesting greater resistance to predictable password formation.

The findings validate that LLMs can serve as predictive models for assessing human password weaknesses, particularly in cultural contexts where PI-driven password construction is common. The Hybrid Similarity Scoring (HSS) approach further confirmed that many generated variants share high semantic and structural resemblance with their originals, reinforcing the notion that Filipino password practices remain highly guessable once contextual information is available.

Recommendations. The study recommends incorporating localized password-checking systems that integrate cultural

and linguistic cues (e.g., Filipino name and word lists) into strength evaluation frameworks. Public awareness campaigns should emphasize the risks of reusing or tweaking passwords based on personal data. Encouraging longer passphrases, non-dictionary structures, and randomized multi-word combinations can significantly improve resilience against LLM-driven and tweaking-based attacks.

Overall, this research establishes a culturally aware, LLM-driven framework for understanding and mitigating credential vulnerabilities. It contributes to both cybersecurity awareness and the development of localized password defense mechanisms tailored for Filipino users. **Future Work**

While this study provides strong evidence for the cultural and semantic sensitivity of LLMs in password analysis, future research can further explore their application as real-time password strength evaluators. Specifically, fine-tuning LLMs with localized Filipino password datasets may enhance their accuracy in detecting PI-based or language-specific vulnerabilities. Integrating LLMs into password creation systems as intelligent evaluators could complement existing tools like zxcvbn, offering users contextual feedback on how personal or linguistic patterns weaken their credentials.

Further work may also include developing an open-source, culturally aware LLM-driven framework for password auditing that balances interpretability, privacy, and computational efficiency. Such advancements can help organizations and end-users better understand password risks beyond entropy scores—by recognizing the human, cultural, and linguistic elements that shape password behavior.

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